Explicit Data-Driven Small-Signal Stability Constrained Optimal Power Flow

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***Abstract*—This paper proposes a data-driven small-signal stability constrained optimal power flow (SSSC-OPF) method with high computational efficiency. Instead of repeating the computational expense small-signal stability analysis via differential and algebraic equations during the iterative OPF process, a computationally cheap surrogate constraint for small-signal stability is developed. To reduce the learning difficulty for small-signal stability boundaries, an efficient sample generation strategy is proposed with sampling space compression. This allows us to use the support vector machine (SVM) with a kernel function to derive the explicit data-driven surrogate constraint for small-signal stability. Penalty factor optimization is proposed to compensate for the error caused by SVM. The learned small-signal stability constraint is embedded into the OPF model for generator control. An examination strategy is also developed to avoid the small-signal instability of re-dispatch caused by the error of the data-driven surrogate model. Comparison results with other model-based and data-driven methods on the IEEE 39-bus and 118-bus systems demonstrate the high computational efficiency and economic benefits of the proposed method.**

***Index Terms*—Optimal power flow, small-signal stability, sensitivity analysis, support vector machine.**

I. Introduction

In POWER system operation, the economic dispatch is calculated by the *optimal power flow* (OPF) algorithms while satisfying the power flow constraints and the equipment operating limits. Then, a security assessment is carried out to ensure that the dispatch results guarantee secure system operation. For large interconnected power systems, the rapid development of ultra-high voltage alternating current transmission [1] and increased penetration of renewable energy resources [2], [3] make it challenging to ensure small-signal stability [4]. A small-signal instability may lead to large-scale blackouts [5], [6]. In the security assessment, when the dispatch result yields a small-signal instability, a re-dispatch is needed. However, the relationship between the small-signal stability and control variables of the power system is established by a set of differential-algebraic equations [8], which are inexplicit and highly nonlinear. Thus, the requirement of small-signal stability is difficult to directly express as an explicit constraint in OPF to obtain the re-dispatch results. Currently, the small-signal instability is solved by the active power adjustment of generators based on the experience of operators in practice [5], [7]. However, this active power adjustment to improve the small-signal stability may lead to large economic losses. The voltage and reactive power are also closely related to the small-signal stability and have a relatively low influence on the generator fuel cost. Therefore, the adjustment of voltage and reactive power should be economically preferable to satisfy small-signal stability requirements [9]. Thus, OPF with computationally tractable small-signal stability constraints should be carefully considered for voltage and reactive power regulation.

Current studies on *small-signal stability constrained OPF* (SSSC-OPF) are briefly reviewed as follows. Because the small-signal stability constraint is inexplicit, these methods are mainly solved by gradient-based optimization algorithms. The gradient is generally obtained by eigenvalue sensitivity analysis and updated at every iteration. In [10], based on a first-order Taylor series expansion, the gradients of small-signal stability constraints are computed by numerical eigenvalue sensitivities. A sequential quadratic programming method combined with gradient sampling is proposed in [11] to improve the convergence and efficiency of SSSC-OPF. In [12] and [13], first-order and second-order eigenvalue sensitivities are calculated by closed-form formulations to update the gradient and Hessian matrix of the primal-dual interior-point algorithm. In [14], the gradient of a small-signal stability constraint is approximately derived by first-order Taylor series expansion and the Hessian matrix is numerically evaluated by perturbing the gradients. In [15], the gradient matrix and Hessian matrix of the small-signal stability constraints are obtained by matrix perturbation theory. In [16], an optimization problem is decomposed into a sequence of subproblems with bounded search regions. These methods modify the active power setpoint, which may compromise the economic efficiency. In [9], the potential for reactive power to help small-signal stability is explored. A gradient-based algorithm for H_2-norm optimization is proposed to calculate the reactive power setpoint. In addition to gradient-based SSSC-OPF, a nonlinear semi-definite programming model is proposed to formulate the

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spectral abscissa constraint as a semi-definite constraint in [17]. Motivated by [18], a novel solution for SSSC-OPF is proposed in [19], which is eigenvalue-analysis independent. In this paper, a convex relaxed formulation of ACOPF [20], [21] is considered to guarantee optimality. This method avoids the limitations of eigenvalue analysis, such as local validity, repeated computation, and nonconvex nature.

The SSSC-OPF method has to obtain a trade-off between cost and stability. If the small-signal stability of a system is set as the constraint, the SSSC-OPF method should have the following features: 1) minimum active power re-dispatch of generators to ensure the economy [22]; 2) being efficient in implementation for practice. The main computational burden of the aforementioned gradient-based methods is to update the gradient and calculate the eigenvalue at every iteration. To avoid repeated eigenvalue calculations, a data-driven method can be used to formulate small-signal stability constraints. Based on the decision tree method, the effect of small-signal stability constraint can be indirectly expressed in the form of capacity limits of critical generators [23], transfer limits of lines [8], or voltage angle limits [24]. An OPF model with these types of constraints is computationally tractable. Data-driven-based methods have higher computational efficiency for online calculation, but two issues remain: 1) Massive amounts of data are needed considering various load patterns, system topologies, and power generations, which cause a considerable burden for data generation and storage [25]. However, only the operation point around the boundary is effective for creating the constraint and most of the data are redundant; 2) The constraints created by the decision tree method cannot explicitly reflect the relationship between the small-signal stability and relative variables. Support vector machine (SVM) has been introduced in transient stability analysis to describe the security boundary [26], [27]. Based on the kernel function of SVM, the relationship between the security indicator (output) and variables (input) can be expressed as an explicit function. SVM is considered as one of the ideal choices to generate the explicit data-driven surrogate constraint for small-signal stability. However, SVM suffers from exceeding time and memory requirements if the training set is very large [28]. This paper proposes an OPF reformulation with an explicit data-driven surrogate constraint for small-signal stability. The contributions are summarized as follows:

1) An efficient sample generation strategy for small-signal stability constraint modeling is proposed by sampling space compression. This helps address the large burden of data generation and storage, significantly enhancing computational efficiency. In particular, the minimum damping ratio of the small-signal stability analysis is set as the output and the voltage magnitudes of generators are set as inputs. The samples that satisfy small-signal stability requirements are considered secure samples, whereas others are insecure samples. The sampling space compression is achieved by the variable reduction and the sampling interval compression. For variable reduction, only the generator voltages that have high sensitivities to the small-signal stability indicator are selected to be adjusted. For sampling interval compression, the acceptable setpoint that close to the boundary and the intial setpoint is approximately calculated by first-order Taylor series expansion. The reduced sampling interval is set at the neighborhood of the acceptable setpoints. In this local sampling space, the non-linear relationship between the small-signal stability and generator voltage can be greatly reduced. This allows us to derive an explicit function for OPF.

2) An OPF reformulation with explicit data-driven surrogate constraint for small-signal stability is developed, where the SVM is adopted to establish the explicit small-signal stability constraint using the proposed sampling strategies. An error compensation strategy with penalty factor optimization is proposed to improve the recall of insecure samples. An examination strategy is proposed to compensate for the error caused by SVM approximation and guarantee the small-signal stability of re-dispatch results.

II. Problem Formulation

The SSSC-OPF is generally expressed below. The objective function is shown as follows:

$$\min \sum_{g\in\Xi} c_{g,2}^{P} P_g^2 + c_{g,1}^{P} P_g + c_{g,0}^{P} \tag{1}$$

where P_g is the active power production of the g^{th} generator; Ξ is the set of generators; $c_{g,2}^{P}$, $c_{g,1}^{P}$ and $c_{g,0}^{P}$ are the coefficients of the cost function of the generator.

The constraints are elaborated below:

1) Nodal Power Balance Equations, $i \in N$:

$$\sum_{g\in i} P_g - P_{i,d} = v_i \sum_{j\in N} v_j \left(G_{ij}\cos\delta_{ij} + B_{ij}\sin\delta_{ij}\right) \tag{2}$$

$$\sum_{g\in i} Q_g - Q_{i,d} = v_i \sum_{j\in N} v_j \left(G_{ij}\sin\delta_{ij} - B_{ij}\cos\delta_{ij}\right) \tag{3}$$

where Q_g is the reactive power production of g^{th} generator; $P_{i,d}$ /$Q_{i,d}$ is the active/reactive power demand at bus *i*; v_i is the voltage magnitude of bus *i*; $\delta_{ij} = \delta_i - \delta_j$, δ_i is the voltage angle of buses *i*; G_{ij} /B_{ij} is the real/imaginary part of Y_{ij} in the admittance matrix; *N* is the set of buses.

2) Branch Flow Constraints, $(i,j) \in K$:

$$P_{ij} = G_{ij}\left(v_i^2 - v_i v_j \cos\delta_{ij}\right) - B_{ij} v_i v_j \sin\delta_{ij} \tag{4}$$

$$Q_{ij} = -B_{ij}\left(v_i^2 - v_i v_j \cos\delta_{ij}\right) - G_{ij} v_i v_j \sin\delta_{ij} \tag{5}$$

$$P_{ij}^2 + Q_{ij}^2 \le S_{ij,\max}^2 \tag{6}$$

$$P_{ji}^2 + Q_{ji}^2 \le S_{ij,\max}^2 \tag{7}$$

where $S_{ij,\max}$ is the apparent power limitation of branch (*i, j*); *K* is the set of branches.

3) *Operational Constraints:*

$$P_g^{\min} \le P_g \le P_g^{\max} \quad g \in \Xi \tag{8}$$

$$Q_g^{\min} \le Q_g \le Q_g^{\max} \quad g \in \Xi \tag{9}$$

$$v_{i,\min} \le v_i \le v_{i,\max} \quad i \in N \tag{10}$$

where $P_g^{\min}/Q_g^{\min}$ and $P_g^{\max}/Q_g^{\max}$are the limits of generator active/reactive power production of g^{th} generator, respectively; $v_{i,\min}$ and $v_{i,\max}$are the limits of voltage at bus *i*.

4) *Small-signal Stability Constraint:*

At a steady-state operating point ($\boldsymbol{x}_{ss}$, $\boldsymbol{u}_{ss}$), the small-signal stability analysis can be described by the following differential and algebraic equations:

$$\dot{\boldsymbol{x}} = \boldsymbol{f}(\boldsymbol{x}, \boldsymbol{u}) \tag{11}$$

$$\boldsymbol{0} = \boldsymbol{g}(\boldsymbol{x}, \boldsymbol{u}) \tag{12}$$

where $\boldsymbol{x}$ is the state vector; $\boldsymbol{u}$ is the input vector; the generator voltage magnitude $\boldsymbol{v}_{g,ss}$ is a part of $\boldsymbol{u}$, $\boldsymbol{v}_{g,ss}\in\boldsymbol{u}$. Equation (11) corresponds to the dynamic model while (12) represents the power flow equations.

The linearizations of (11) and (12) at the equilibrium point $\boldsymbol{x}_{ss}$ and $\boldsymbol{u}_{ss}$ are shown as follows:

$$\Delta\dot{\boldsymbol{x}} = \boldsymbol{A}\Delta\boldsymbol{x} + \boldsymbol{B}\Delta\boldsymbol{u} \tag{13}$$

$$\boldsymbol{0} = \boldsymbol{C}\Delta\boldsymbol{x} + \boldsymbol{D}\Delta\boldsymbol{u} \tag{14}$$

where $\mathbf{A} = \frac{\partial \mathbf{f}}{\partial \mathbf{x}}|_{\mathbf{x}=\mathbf{x}_{ss}}$, $\mathbf{B} = \frac{\partial \mathbf{f}}{\partial \mathbf{u}}|_{\mathbf{u}=\mathbf{u}_{ss}}$, $\mathbf{C} = \frac{\partial \mathbf{g}}{\partial \mathbf{x}}|_{\mathbf{x}=\mathbf{x}_{ss}}$, $\mathbf{D} = \frac{\partial \mathbf{g}}{\partial \mathbf{u}}|_{\mathbf{u}=\mathbf{u}_{ss}}$; $\boldsymbol{B}$ and $\boldsymbol{D}$ are the functions that involve the steady-state value of generator voltage $\boldsymbol{v}_{g,ss}$, $\boldsymbol{v}_{g,ss}\in\boldsymbol{u}_0$.

Combining (13) and (14), and eliminating $\Delta\boldsymbol{u}$, the system state equations can be expressed as:

$$\Delta\dot{\boldsymbol{x}} = \boldsymbol{A}_s\Delta\boldsymbol{x} \tag{15}$$

where $\mathbf{A}_s = (\mathbf{A} - \mathbf{B}\mathbf{D}^{-1}\mathbf{C})$ is the state matrix; $\boldsymbol{A}_s$ is affected by the steady-state value of generator voltage $\boldsymbol{v}_{g,ss}$, $\boldsymbol{v}_{g,ss}\in\boldsymbol{u}_{ss}$. The eigenvalues of $\boldsymbol{A}_s$ and their damping ratios are expressed as:

$$\lambda = \sigma \pm j\omega \tag{16}$$

$$\zeta = \frac{-\sigma}{\sqrt{\sigma^2 + \omega^2}} \tag{17}$$

The eigenvalues $\lambda = \sigma \pm j\omega$ are considered as the indicators for the small-signal stability.σrepresents the damping of oscillation and ω is the frequency of oscillation. According to the characteristics of eigenvalues, the damping ratio ζ is a comprehensive indicator for oscillation, which reflects the performance of the decay of oscillation. It should be large enough to ensure small-signal stability. Generally, the damping ratio should be beyond the critical damping ratio ζ_c (normally set as 3%–5%) [16].

The minimum damping ratio can be selected as the small-signal stability indicator [29]. Regarding the generator voltage as the control variables to improve the damping ratio, according to (13)–(17), the minimum damping ratio ζ_m can be regarded as a function of $\boldsymbol{v}_{g,ss}$:

$$\zeta_m = \min(\boldsymbol{\zeta}) = h(\boldsymbol{v}_{g,ss}) \tag{18}$$

where $h()$ refers to the relationship between ζ_m and $\boldsymbol{v}_{g,ss}$, which is an implicit function of $\boldsymbol{v}_{g,ss}$.

If $\zeta_m>\zeta_c$, the corresponding setpoint can be considered as secure and vice visa. If $\zeta_m = \zeta_c$, the corresponding setpoint is on the boundary. Thus, the small-signal stability constraint can be expressed as:

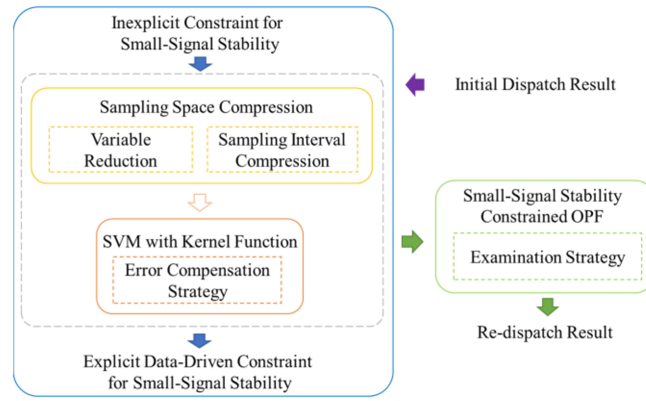


Fig. 1. The proposed explicit data-driven small-signal stability constrained optimal power flow method.

$$\zeta_m = h(\boldsymbol{v}_{g,ss}) \geq \zeta_c \tag{19}$$

In summary, the SSSC-OPF can be established by treating (1) as an objective function with constraints (2)-(9) and (18).

Proposed method: According to (15)-(17), $h(\boldsymbol{v}_{g,ss})$ is an implicit function of $\boldsymbol{v}_{g,ss}$. Based on the mapping relationship between ζ_m and $\boldsymbol{v}_{g,ss}$, the explicit expression of $h(\boldsymbol{v}_{g,ss})$ can be learned by the data-driven method. However, the expression of $h(\boldsymbol{v}_{g,ss})$ is non-linear. It is also difficult to accurately express (19) in a computationally tractable form using a small volume of data. To address this issue, the main structure of the proposed method is shown in Fig. 1. The dimension and size of the sampling space are decreased by generating the samples closely around the boundary. The learning difficulty can be effectively reduced. Based on the sensitivity analysis, only the generator voltages with large sensitivities to ζ_m are selected to be adjusted to reduce the dimension of the sampling space. Furthermore, the sampling interval of each variable is reduced to compress the sampling space. The compressed sampling interval must include the boundary and be as close as possible to the initial setpoint to ensure the economy. Therefore, we first find a certain critical stability point that is closest to the initial point and set a sampling interval around it. In this lower dimension space, the small-signal stability constraint (19) can be effectively learned by SVM with a kernel function to establish an explicit data-driven surrogate constraint for small-signal stability. This constraint can be conveniently embedded into the OPF model for solutions. An error examination and compensation strategy are proposed to mitigate the learning error of SVM.

The proposed method avoids repeated eigenvalue calculations during the optimal process and shows the desired computational efficiency. Different from the existing data-driven-based methods, which need to consider various load patterns, system topology, and power generation to satisfy the sampling diversity, the proposed method generates samples online for a certain unstable setpoint. This avoids the considerable burden of data generation and storage.

III. An Efficient Sample Generation Strategy

A. Motivation

Generally, the denser the samples around the boundary, the more accurate the formulated boundary is. The distance between samples *d* is mainly affected by the size of sampling space *S* and the size of datasets *n*. Their relationship can be described by the following equation.

$$S \propto d, n \tag{20}$$

Thus, if we want to obtain a small *d*, *S* should be small. In addition, to ensure the efficiency of sample generation, we expect to formulate the boundary with a small number of samples, which also means that *S* should be small.

Then the size of sampling space *S* is determined by the sampling dimension, which is determined by the number of variables *m* and the sampling interval ε_1 for each variable. Their relationship is described as:

$$S \propto m, \varepsilon_1 \tag{21}$$

Thus, variable reduction and sampling interval compression are proposed to compress the sampling space. Furthermore, the shape of the small-signal stability boundary is complex and non-linear, which is hard to be described as an explicit constraint and integrated into optimization. With the compression of the sampling space, the boundary restricted to a small area can be regarded as quasi-linear. The detailed strategy to compress the sampling space is described as follows.

B. Variable Reduction With Sensitivity Analysis

To compress the sampling space, the first step is to reduce the dimension of the sampling space. A strategy to decrease the number of variables is proposed in this section. Both generator active output and generator voltage have impacts on small-signal stability. However, the economy of a power system is closely related to generator active output. Hence, the voltage magnitude is selected as the adjustment variable to help avoid the conflict between stability and economy. Considering the economy, only the generator voltage magnitudes, which generally have little relationship with the generator active output, are chosen as variables in this paper. For a large-scale system, the number of generators can be large. According to the simulations, only a small number of generator voltage magnitudes have a significant impact on the small-signal stability. This means that the generator voltage magnitudes with small sensitivity to the small-signal stability contribute less to improving the stability. Thus, only generator voltages with large sensitivities are considered. The variable selection strategy is shown as follows.

$$s_{0,j} = \frac{\partial \zeta_m}{\partial v_{g0,j}} \geq s_l \tag{22}$$

where s_l is a sensitivity threshold to select the variables. We provide an instructive strategy to set s_l. The voltage magnitudes are arranged in descending order of sensitivity. The voltage magnitudes with the largest sensitivity and others, whose sensitivities are on the same order of magnitude, should be selected. The value of s_l needs to satisfy the above rule. $v_{g0,j}$ is the voltage magnitude obtained by the OPF without the small-signal stability constraint; $s_{0,j}$ is the sensitivity of ζ_m to $v_{g0,j}$ [30], i.e.,:

$$\begin{aligned}\frac{\partial \zeta_m}{\partial v_{g0,j}} &= \frac{\partial \left(\frac{-\sigma_m}{\sqrt{\sigma_m^2+\omega_m^2}}\right)}{\partial v_{g0,j}} \\ &= \frac{-\omega_m^2 \mathrm{Re}\left(\frac{\partial \lambda_m}{\partial v_{g0,j}}\right) + \sigma_m \omega_m \mathrm{Im}\left(\frac{\partial \lambda_m}{\partial v_{g0,j}}\right)}{|\lambda_m^3|} \quad j \in \boldsymbol{A}_v\end{aligned} \tag{23}$$

where $\lambda_m = \sigma_m \pm j\omega_m$ is the eigenvalue corresponding to ζ_m; $\boldsymbol{A}_v$ is the set of generator voltages to be adjusted. The sensitivities $\partial\lambda_m/\partial v_{g0,j}$ can be obtained via:

$$\frac{\partial \lambda_m}{\partial v_{g0,j}} = \frac{\boldsymbol{v}_m^T \frac{\partial \boldsymbol{A}_s}{\partial v_{g0,j}} \boldsymbol{u}_m}{\boldsymbol{v}_m^T \boldsymbol{u}_m} \quad j \in \boldsymbol{A}_v \tag{24}$$

where $\boldsymbol{v}_m$ and $\boldsymbol{u}_m$ are the left and right eigenvectors of eigenvalue λ_m respectively, yielding [11]:

$$\begin{aligned}\frac{\partial \boldsymbol{A}_s}{\partial v_{g0,j}} =& \frac{\partial \boldsymbol{A}}{\partial v_{g0,j}} - \frac{\partial \boldsymbol{B}}{\partial v_{g0,j}} \boldsymbol{D}^{-1}\boldsymbol{C} \\ &+ \boldsymbol{B}\boldsymbol{D}^{-1}\frac{\partial \boldsymbol{D}}{\partial v_{g0,j}}\boldsymbol{D}^{-1}\boldsymbol{C} - \boldsymbol{B}\boldsymbol{D}^{-1}\frac{\partial \boldsymbol{C}}{\partial v_{g0,j}} \quad j \in \boldsymbol{A}_v\end{aligned} \tag{25}$$

Based on a first-order Taylor series expansion at $\boldsymbol{v}_{g0,j}$, the relationship between ζ_m and selected generator voltage $\boldsymbol{v}_g$ can be expressed as:

$$\zeta_m = \zeta_{m0} + \sum_{j \in \boldsymbol{A}_v} \frac{\partial \zeta_m}{\partial v_{g0,j}} (v_{g,j} - v_{g0,j}) + R(\boldsymbol{v}_g) \tag{26}$$

where ζ_{m0} is the corresponding minimum damping ratio of v_{g0}; $R(\boldsymbol{v}_{\mathrm{g}})$ is the remainder of the first-order Taylor series expansion.

To ensure small-signal stability, ζ_m should be larger than ζ_{m0} after voltage adjustment. To achieve this, each voltage adjustment should satisfy:

$$\frac{\partial \zeta_m}{\partial v_{g0,j}} (v_{g,j} - v_{g0,j}) > 0 \quad j \in \boldsymbol{A}_v \tag{27}$$

The generator voltage should satisfy its limits [$v_{g,j\ \min}$, $v_{g,j\ \max}$]. The largest voltage adjustment can be expressed as:

$$\Delta v_{g,j\,\max} = \begin{cases} v_{g,j\,\max} - v_{g0,j} & if \frac{\partial \zeta_m}{\partial v_{g0,j}} > 0 \\ v_{g,j\,\min} - v_{g0,j} & if \frac{\partial \zeta_m}{\partial v_{g0,j}} < 0 \end{cases} \quad j \in \boldsymbol{A}_v \tag{28}$$

Ignoring $R(\boldsymbol{v}_g)$ and relying on (23), (26) and (28), the selected set of generator voltages $\boldsymbol{A}_v$ should at least satisfy:

$$\zeta_{m0} + \sum_{j \in \boldsymbol{A}_v} \frac{\partial \zeta_m}{\partial v_{g0,j}} \Delta v_{g,j\,\max} \geq \zeta_c \tag{29}$$

If the selected variables cannot satisfy (29), more voltage magnitudes should be selected into $\boldsymbol{A}_v$ according to the sensitivity order until (29) is satisfied. If (29) is still infeasible when all the voltage magnitudes are selected, additional variables, which may sacrifice more economy should be included. This issue is not solved in this paper and further studies will be conducted.

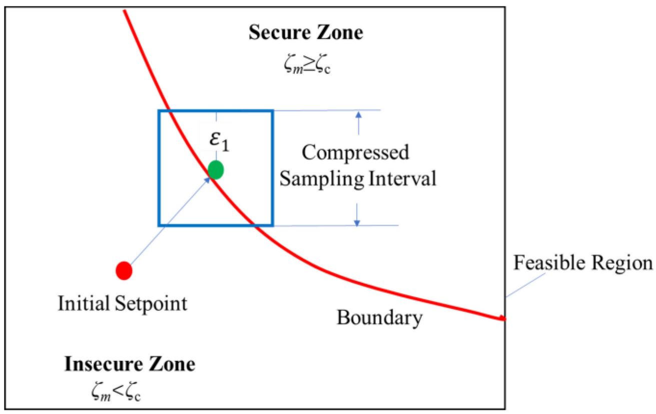


Fig. 2. The proposed sampling interval compression method.

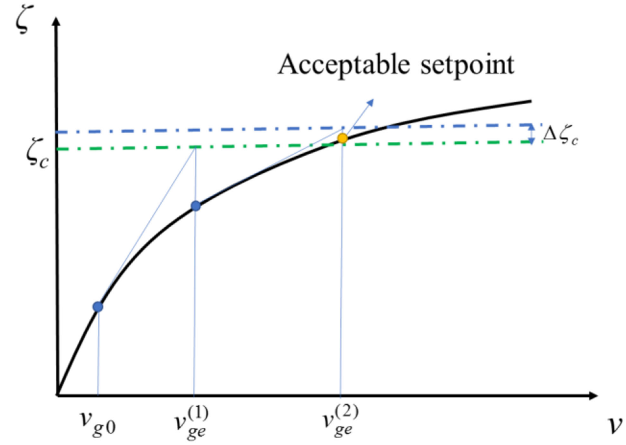


Fig. 3. The iteration process to find an acceptable setpoint.

C. Compression of Sampling Interval

In the sampling space after dimension reduction, the sampling interval is compressed to further simplify the shape of the boundary to be studied. The sampling space after the sampling interval compression should satisfy the following requirements: 1) it involves part of the boundary; 2) the samples should be balanced; 3) it is close to the initial dispatch setpoint.

To achieve this outcome, a setpoint with ensured small-signal stability that is close to the initial dispatch setpoint should be identified (as illustrated by the green point in Fig. 2). Based on (26), this setpoint is found by the optimal model (30) as follows:

$$\begin{aligned} &\min \sum_{j\in \boldsymbol{A}_v} (\Delta v_{ge,j})^2 \\ &\text{s.t.} \;\; \zeta_c = \zeta_{m0} + \sum_{j\in \boldsymbol{A}_v} s_{0,j}\Delta v_{ge,j} \end{aligned} \tag{30}$$

where $\Delta v_{ge,j} = v_{ge,j} - v_{g0,j}$; $v_{ge,j}$ is the j^{th} voltage magnitude of the expected secure setpoint, $j \in \mathbf{A}_v$.

Based on the exterior point method, the auxiliary function (30) can be expressed as:

$$\begin{aligned} F(\Delta \boldsymbol{v}_{ge}, \mu) = &\sum_{j\in \boldsymbol{A}_v} (\Delta v_{ge,j})^2 \\ &+ \frac{\mu}{2}\left((\zeta_{m0} - \zeta_c) + \sum_{j\in \boldsymbol{A}_v} s_{0,j}\Delta v_{ge,j}\right)^2 \end{aligned} \tag{31}$$

where μ is the penalty parameter, which is a sufficiently large positive number; $\Delta \mathbf{v}_{ge}$ is the set of $\Delta v_{ge,j}$, $j \in \mathbf{A}_v$.

According to the optimality condition $\nabla F(\Delta \mathbf{v}_{ge}, \mu) = 0$, $\Delta \mathbf{v}_{ge}$ can be solved by:

$$\begin{bmatrix} \Delta v_{ge,1} \\ \Delta v_{ge,2} \\ \vdots \\ \Delta v_{ge,n_a} \end{bmatrix} = \begin{bmatrix} 2{+}\mu s_{0,1}^2 & \mu s_{0,1}s_{0,2} & \cdots & \mu s_{0,1}s_{0,n_a} \\ \mu s_{0,2}s_{0,1} & 2{+}\mu s_{0,2}^2 & \cdots & \mu s_{0,2}s_{0,n_a} \\ \vdots & \vdots & & \vdots \\ \mu s_{0,n_a}s_{0,1} & \mu s_{0,n_a}s_{0,2} & & 2{+}\mu s_{0,n_a}^2 \end{bmatrix}^{-1} \times \begin{bmatrix} \mu s_{0,1}(\zeta_c - \zeta_{m0}) \\ \mu s_{0,2}(\zeta_c - \zeta_{m0}) \\ \vdots \\ \mu s_{0,n_a}(\zeta_c - \zeta_{m0}) \end{bmatrix} \tag{32}$$

where n_a is the total number of generator voltages to be adjusted.

The generator voltage of the expected secure setpoint is shown as:

$$v_{ge,j} = v_{g0,j} + \Delta v_{ge,j} \quad j \in \boldsymbol{A}_v \tag{33}$$

Considering that $\Delta \mathbf{v}_{ge}$ may be large, the relationship between the damping ratio and voltage magnitude is nonlinear. (30) is based on the first-order Taylor series expansion, which is linear. Therefore, the stable setpoint is difficult to directly obtain by (30)–(33). An iteration based on the Newton method to find the expected secure setpoint is proposed. The basic concept of this iteration is shown in Fig. 3.

To this end, (32) and (33) are rewritten as

$$\begin{bmatrix} \Delta v_{ge,1}^{(k+1)} \\ \Delta v_{ge,2}^{(k+1)} \\ \vdots \\ \Delta v_{ge,n_a}^{(k+1)} \end{bmatrix} = \begin{bmatrix} 2{+}\mu\left(s_1^{(k)}\right)^2 & \mu s_1^{(k)} s_2^{(k)} & \cdots & \mu s_1^{(k)} s_{n_a}^{(k)} \\ \mu s_2^{(k)} s_1^{(k)} & 2{+}\mu\left(s_2^{(k)}\right)^2 & \cdots & \mu s_2^{(k)} s_{n_a}^{(k)} \\ \vdots & \vdots & & \vdots \\ \mu s_{n_a}^{(k)} s_1^{(k)} & \mu s_{n_a}^{(k)} s_2^{(k)} & & 2{+}\mu\left(s_{n_a}^{(k)}\right)^2 \end{bmatrix}^{-1} \times \begin{bmatrix} \mu s_1^{(k)}\left(\zeta_c^{(k)} - \zeta_{me}^{(k)}\right) \\ \mu s_2^{(k)}\left(\zeta_c^{(k)} - \zeta_{me}^{(k)}\right) \\ \vdots \\ \mu s_{n_a}^{(k)}\left(\zeta_c^{(k)} - \zeta_{me}^{(k)}\right) \end{bmatrix} \tag{34}$$

$$v_{ge,j}^{(k+1)} = v_{ge,j}^{(k)} + \Delta v_{ge,j}^{(k+1)} \quad j \in \boldsymbol{A}_v \tag{35}$$

$$s_j^{(k)} = \frac{\partial \zeta_m^{(k)}}{\partial v_{ge,j}^{(k)}} \quad j \in \boldsymbol{A}_v \tag{36}$$

where $\zeta_{me}^{(k)}$ is the minimum damping ratio regarding $\mathbf{v}_{ge}^{(k)}$.

The algorithm to identify secure setpoints is as follows:

Algorithm 1: Acceptable Setpoint Search Strategy.

0. Set $v_{ge}^{(k)}=v_{g0}$, $\zeta_{me}^{(k)}=\zeta_{m0}$, $\zeta_c^{(k)}=\zeta_c$, $s^{(k)}=s_0 k\leftarrow 0$
1. Calculate $v_{ge,j}^{(k+1)}$ by (34) and (35)
2. Substitute $v_{ge,j}^{(k+1)}$ into modal analysis and calculate $\zeta_{me}^{(k+1)}$
3. If $\zeta_{me}^{(k+1)}>\zeta_c^{(k)}$, terminate the algorithm and set $v_{ge,j}^{(k+1)}$ as the objective secure setpoint $v_{gb,j}=v_{ge,j}^{(k+1)}$
4. If $k>k_{\max}$, terminate the algorithm
5. Calculate all the sensitivities of voltage magnitudes by (23). If one other $s_j^{(k+1)}\geq s_l$, this voltage magnitude should be added into $\boldsymbol{A}_v$. $s_j^{(k+1)}, j\in \mathbf{A}_v$ is obtained
6. Set $\zeta_c^{(k+1)}=\zeta_c^{(k)}+\Delta\zeta_c$, $\Delta\zeta_c>0$. $k\leftarrow k+1$ and go to Step 1

Different from the optimization process in [10], this setpoint does not need to be optimal. Instead, the optimal solution is searched from the setpoint using a few iteration steps.It should be noted that Step 4 of the algorithm and (29) are used together as the applicable condition for the proposed method. $\Delta\zeta_c$ is added at each step to accelerate the convergence rate. The sampling interval is set at the neighborhood of $v_{ge,j}$, which is $[v_{ge,j}-\varepsilon_1, v_{ge,j}+\varepsilon_1]$. Based on (20) and (21), the sampling interval is determined as:

$$\varepsilon_1 = D\frac{n}{m} \tag{37}$$

where *D* is a coefficient to qualitatively describe the distance *d*.

The samples are generated via:

$$v_{gs,j}\in[v_{ge,j}-\varepsilon_1, v_{ge,j}+\varepsilon_1] \quad j\in \boldsymbol{A}_v \tag{38}$$

Thus, the limits of the adjusted generator voltage magnitude should be modified as:

$$v_{ge,j}-\varepsilon_1\leq v_{g,j}\leq v_{ge,j}+\varepsilon_1 \quad j\in \boldsymbol{A}_v \tag{39}$$

where $v_{gs,j}$ represents the generated generator voltage samples.

Based on (38), the samples $\{(\mathbf{v}_{gs},\zeta_{ms})\}_{s=1}^{n}$can be obtained by the Latin hypercube sampling method, where ζ_{ms} is the minimum damping ratio corresponding to $\boldsymbol{v}_{gs}$.

IV. OPF With Data-Driven Surrogate Small-Signal Stability Constraint

Based on the samples generated in Section II, the data-driven surrogate constraint for small-signal stability in the form of generator voltages is formulated and further embedded in the OPF model. To establish the data-driven small-signal stability constraint, the small-signal stability boundary should be identified first. If $\zeta_s>\zeta_c$, this sample is secure; otherwise, it is labeled as insecure. This is a typical binary classification discrimination problem. In this paper, we select SVM [31] to formulate the small-signal stability constraints, as it can describe the small-signal boundary with an explicit kernel function even with small numbers of samples. In addition, taking advantage of the SVM learning characteristics, the selected data-driven method is expected to improve the recall of the insecure samples by setting a tough penalty of misclassifying the insecure samples.

A. Data-Driven Small-Signal Stability Constraint

For the SVM, the datasets $\{(\mathbf{x}_i,y_i)\}_{i=1}^{n}$ for the binary classification learning problem consist of objects labeled with $y_i=+1$ or $y_i=-1$. In this paper, the input for SVM is the adjusted voltage magnitude selected in Section.III.A. The secure sample is labeled as $(\mathbf{v}_{gs},+1)$ while the insecure sample is labeled as $(\mathbf{v}_{gs},-1)$. A brief introduction of SVM is shown below.

The reason for choosing the linear kernel function is explained as follows. Because the boundary is restricted to a small area, the relationship between the minimum damping ratio and voltage magnitudes can be regarded as quasi-linear. Additionally, the linear form of the kernel function can address the computational challenges by avoiding involving additional nonconvexity. The discriminant function of the SVM can be expressed as:

$$f(\boldsymbol{x})=\boldsymbol{w}^{\mathrm{T}}\boldsymbol{x}+b \tag{40}$$

where *w* is the weight vector; *b* is the bias. The decision boundary of the classifier can be expressed as a hypersurface:

$$\left\{\boldsymbol{x}: f(\boldsymbol{x})=\boldsymbol{w}^{\mathrm{T}}\boldsymbol{x}+b=0\right\} \tag{41}$$

To obtain the values of *w* and *b*, based on soft margin SVM, the optimization problem can be established as:

$$\begin{aligned}&\min\quad \frac{1}{2}\|\boldsymbol{w}\|^2+C\sum_{i=1}^{n}\gamma_i\\ &\text{s.t}\begin{cases} y_i\left(\boldsymbol{w}^{\mathrm{T}}\boldsymbol{x}_i+b\right)\geq 1-\gamma_i\\ \gamma_i\geq 0\end{cases}\quad i=1,2,\ldots,n\end{aligned} \tag{42}$$

where $\|\mathbf{w}\|$ is the norm of the vector *w* and its length is given by $\sqrt{\mathbf{w}^{\mathrm{T}}\mathbf{w}}$;$\gamma_i\geq 0$represents slack variables; *C* is the penalty factor to penalize the misclassification and margin errors.

Using a Lagrange multiplier method, we can obtain the dual formulation, which is expressed in terms of variables α_i [32]:

$$\begin{aligned}&\min\quad \sum_{i=1}^{n}\alpha_i-\tfrac{1}{2}\sum_{i=1}^{n}\sum_{j=1}^{n}y_iy_j\alpha_i\alpha_j\boldsymbol{x}_i^{\mathrm{T}}\boldsymbol{x}_j\\ &s.t\begin{cases}\sum_{j=1}^{n}y_i\alpha_i=0\\ 0\leq\alpha_i\leq C\end{cases}\quad i=1,2,\ldots,n\end{aligned} \tag{43}$$

The parameters *w* and *b* can be calculated by:

$$\begin{cases}\boldsymbol{w}^*=\sum_{i=1}^{n}\alpha_i^*y_i\boldsymbol{x}_i\\ b^*=-\tfrac{1}{2}\boldsymbol{w}^*(\boldsymbol{x}_++\boldsymbol{x}_-)\end{cases} \tag{44}$$

where α_i^* is the optimal solution of (43); $\boldsymbol{x}_+$/$\boldsymbol{x}_-$ are the closest points to the hyperplane among the positive/negative examples.

The secure samples are labeled as $(\mathbf{v}_{gs},+1)$, and the setpoint should be classified as a positive set. Therefore, the proposed data-driven surrogate constraint can be expressed as:

$$\boldsymbol{w}^{\mathrm{T}}\boldsymbol{v}_{ga}+b\geq 0 \tag{45}$$

where $\boldsymbol{v}_{ga}$ is the vector of $v_{g,j}$, $j\in\boldsymbol{A}_v$.

Algorithm 2: Set the Value of C_+ and C_-

0. Set $C_+^{(l)}=1$, $C_-^{(l)}=1$, $l\leftarrow 0$
1. Conduct SVM and calculate R
2. If $R_{\cdot thr}\leq R$, terminate the algorithm
3. If $l>l_{\max}$, terminate the algorithm
4. $C_-^{(l+1)}=C_-^{(l)}+\Delta C_-$, $\Delta C_->0$, $l\leftarrow l+1$ and go to Step 1

B. Compensation Strategy for Classification Error

Errors in the SVM are inevitable. For small-signal stability, there may be a case where the insecure sample is estimated as secure. This should be avoided because of security concerns. In contrast, a case where the secure samples are estimated to be insecure may induce additional costs but do not threaten system security. The recall of the insecure samples R is defined as follows:

$$R=\frac{TP}{TP+FN} \tag{46}$$

where TP is the number of true insecure samples; FN is the number of insecure samples determined as secure.

To improve the recall of the insecure samples, R should be as large as possible. The penalty of misclassifying the insecure samples should be tough in (42), yielding

$$\min\ \frac{1}{2}\|\boldsymbol{w}\|^2+C_+\sum_{i\in I_+}\gamma_i+C_-\sum_{i\in I_-}\gamma_i$$

$$\text{s.t}\begin{cases} y_i\left(\boldsymbol{w}^{\mathrm{T}}\boldsymbol{x}_i+b\right)\geq 1-\gamma_i \\ \gamma_i\geq 0\end{cases}\quad i=1,2,\ldots,n \tag{47}$$

where C_+/C_- are the soft-margin constants for the secure/insecure samples, I_+/I_- are the sets of secure/insecure samples and $C_+\leq C_-$. The values of C_+/C_- are calculated by Algorithm 2.

The proposed compensation strategy can decrease the probability of misclassifying insecure samples, leading to improved capability of the proposed method to ensure the system's small-signal stability.

C. Solution of Data-Driven SSSC-OPF

With (1)–(10), (39) and (45), the proposed data-driven small-SSSC-OPF model is established. This model can be directly solved by non-linear optimization solvers, such as the interior-point solver and ipopt solver [33]. Thus, the proposed method is much more efficient than the traditional iterative method.

To ensure the small-signal stability of the re-dispatch result, an examination strategy is proposed. If ζ_m still does not reach ζ_c with the above step, the constraint (45) should be set more strictly, i.e.,

$$\boldsymbol{w}^{\mathrm{T}}\boldsymbol{v}_{ga}+b\geq m\tau \tag{48}$$

where m is the number of repeated calculation of data-driven SSSC-OPF; τ is the step size,$\tau>0$.

The constraint (48) is substituted into the proposed method to re-execute the optimization process until $\zeta_m\geq\zeta_c$. This process is needed in very few cases because the accuracy of the learned small-signal stability boundary is desirable and a compensation strategy is used to further ensure security.

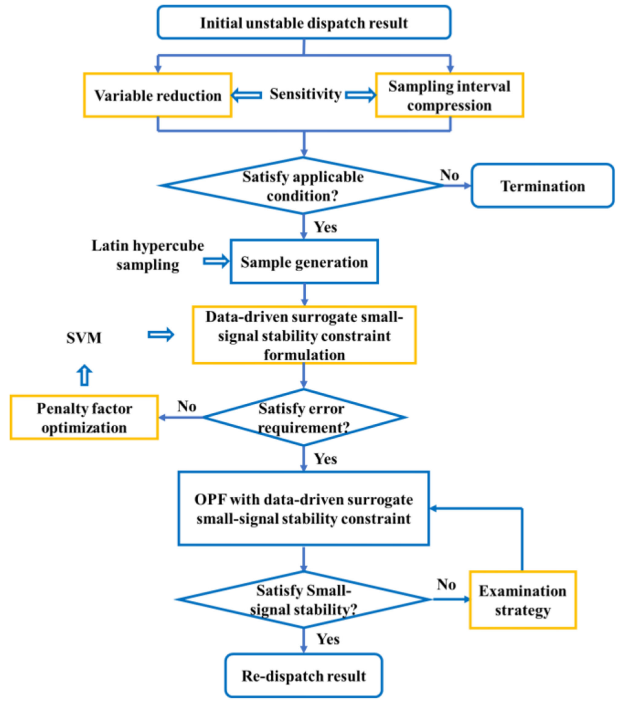


Fig. 4. The framework of the proposed data-driven OPF method.

D. Framework of the Proposed Method

The framework of the proposed method is summarized in Fig. 4. In this paper, the initial dispatch result is obtained by ACOPF without a stability constraint. If the initial dispatch result is judged as small-signal instability, the proposed method is conducted to find a stable re-dispatch result. Based on the initial unstable dispatch result and sensitivity analysis, variable reduction and sampling interval compression are proposed to enhance computational efficiency. If the applicable conditions derived based on Step 4 of Algorithm 1 and (29) are satisfied, the sample is generated by the Latin hypercube sampling approach. Then, the generated samples are trained by SVM to formulate the data-driven surrogate small-signal stability constraint. To decrease the probability of misclassifying insecure samples, penalty factor optimization for SVM is proposed. Finally, the proposed data-driven surrogate small-signal stability constraint is integrated into OPF to obtain the re-dispatch result. To ensure the small-signal stability of the re-dispatch result, an examination strategy is proposed. If the result of the proposed SSSC-OPF is still unstable, the limit of constraint (48) is strengthened until a stable re-dispatch result is obtained.

V. Simulation Results

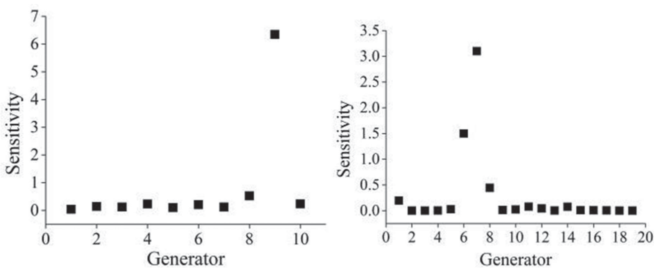
Modified IEEE 39-bus and 118-bus systems are used to illustrate the effectiveness of the proposed method. Descriptions of these two systems can be found in [34]. The 39-bus

TABLE I
THE PARAMETER SETTING

Parameter	k_{max}	$\Delta\zeta_c$	ε_1	n	ΔC_{-}	R	l_{max}	τ
Value	10	0.001	0.01	100	0.5	100%	50	1

TABLE II
THE INITIAL DISPATCH RESULTS WITHOUT CONSIDERING SMALL-SIGNAL STABILITY CONSTRAINTS

Test system	ζ_c(%)	ζ_{m0}(%)	Adjusted voltage	Sensitivity	$\Delta v_{g\max}$ (p.u.)
IEEE 39-bus system	3	-1.12	v_{38}(Gen9)	6.34	0.105
IEEE 118-bus system	5	-3.02	v_{25}(Gen6)	−1.50	0.120
			v_{26}(Gen7)	3.10	0.096



(a) IEEE 39-bus system (b) IEEE 118-bus system

Fig. 5. The distribution of sensitivity.

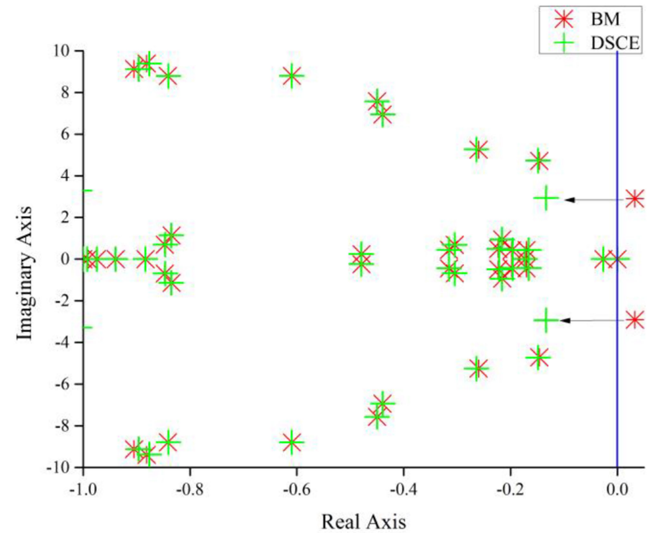


Fig. 6. Eigenvalue trajectory.

TABLE III
ACCURACIES OF SVM

Test system	Training set	Test set
IEEE 39-bus system	100%	100%
IEEE 118-bus system	84.28%	93.33%

system has 10 generators while the 118-bus system has 54 synchronous machines, including 19 generators, 20 synchronous compensators used for reactive power support, and 15 motors. The dynamic models of the generator, excitation system, and governor are described by the built-in dynamic models of DIgSILENT/PowerFactory, see [35]. Excitation systems are modeled by the IEEE Type 1 exciters and governors are modeled by the IEEE Type 1 speed-governing model. The specific data of the test systems can be found in [36]. We use a linear kernel of SVM to discuss the effectiveness of the proposed method. The optimization problem is solved by Matpower. The parameters are shown in Table I.

The initial dispatch result without considering small-signal stability is shown in Table II and it can be seen that the minimum damping ratios ζ_{m0} of these two systems are below their critical damping ratio ζ_c. The distribution of sensitivities is shown in Fig. 5. Only one or two generator voltage magnitudes have a significant impact on the small-signal stability in the test cases. According to Fig. 5, the selected generator voltages for each test system are shown in Table II.

Based on Algorithm 1, the estimated secure setpoint $v_{ge,j}$ for the IEEE 39-bus system is $v_{38} = 0.962$ p.u. and for the IEEE 118-bus system, $v_{25} = 1.054$ p.u., $v_{26} = 1.009$ p.u.. The sampling space is set at the neighborhood of the estimated secure setpoint. The iteration of Algorithm 1 for the IEEE 39-bus system is $k = 1$ and for the IEEE 118-bus system, $k = 3$, which all satisfy $k \leq k_{\max}$. The number of samples is set as $n = 100$. 70% of the samples are the training samples and 30% are testing samples. The accuracy of SVM is demonstrated in Table III, which shows excellent performance after implementing the proposed strategies.

A. Effectiveness of the Proposed Method

To analyze the effectiveness of the proposed method, we consider the following three comparison methods:

BM: Benchmark, where the initial dispatch result is obtained by the standard OPF without small-signal stability constraints.

SCI: SSSC-OPF based on eigenvalue sensitivity iteration [10].

DSC: Proposed explicit data-driven SSSC-OPF without error compensation and examination strategy.

DSCE: Proposed explicit data-driven SSSC-OPF with error compensation and examination strategy

The simulation results for different methods are shown in Table IV. The minimum damping ratios obtained by SCI and DSCE both reach ζ_c but the economic sacrifice of SCI is larger than that of DSCE. For the IEEE 39-bus system, SCI spends \$1912/h to achieve small-signal stability, while DSCE only needs \$7/h. For the IEEE 118-bus system, the economic loss of SCI is almost 6 times larger as compared to DSCE to satisfy the small-signal stability. For the error of SVM, the damping ratio of DSC is just 4.85% in the IEEE118-bus system, which cannot satisfy $\zeta_c = 5\%$.

The accuracy of the sample space reduction is then tested. We use the solution obtained from the enumerative search in the complete space of selected voltages in Table II as the benchmark.

TABLE IV
SIMULATION RESULTS OF DIFFERENT METHODS

Test system	BM		SCI		DSC		DSCE	
	Generation Cost($/h)	ζ_m(%)	Generation Cost($/h)	ζ_m(%)	Generation Cost($/h)	ζ_m(%)	Generation Cost($/h)	ζ_m(%)
IEEE 39-bus system	213016	-1.12	214926	3.13	213021	3.13	213021	3.13
IEEE 118-bus system	170722	-3.02	172366	5.04	170943	4.85	170974	5.27

TABLE V
THE PERFORMACE OF THE SMAPLE SPACE REDUCTION

Test system	Generation Cost($/h)		
	Complete space	Reduced space	Gap
IEEE 39-bus system	213017	213021	0.0017%
IEEE 118-bus system	170946	170974	0.0158%

TABLE VI
GENERATOR DISPATCH RESULTS OF BM, SCI AND DSCE FOR IEEE 39-BUS SYSTEM

Test system	P_G(MW)			V_G (p.u.)		
	BM	SCI	DSCE	BM	SCI	DSCE
Gen 1	965.6	972.8	965.3	1.060	1.060	1.060
Gen 2	595.0	571.7	595.0	0.940	0.940	0.940
Gen 3	445.8	451.0	446.0	0.954	0.954	0.954
Gen 4	380.6	385.5	380.6	0.944	0.944	0.943
Gen 5	510.0	507.7	510.0	0.958	0.958	0.957
Gen 6	546.4	553.0	546.5	0.994	0.994	0.992
Gen 7	435.1	440.0	435.2	1.021	1.021	1.020
Gen 8	378.1	382.9	378.3	1.000	1.000	0.998
Gen 9	1180.0	1164.8	1179.0	0.955	0.955	0.963
Gen10	98.0	103.6	98.0	0.940	0.940	0.940

TABLE VII
THE RECALL OF THE INSECURE SAMPLES R

Test system	IEEE 39-bus system	IEEE 118-bus system
R	100%	95.08%

The results are shown in Table V. It can be observed that the sample space reduction brings a limited influence on accuracy.

The simulation results demonstrate that with the proposed small-signal stability constraint, the minimum damping ratio can be improved to a secure level and the generation dispatch cost is slightly affected. Furthermore, with the proposed sample generation strategy, only a small sample size is sufficient to ensure the effectiveness of the proposed method.

Taking the IEEE 39-bus system as an example, the generator outputs and voltages by BM-DSCE are shown in Table VI. For DSCE, the voltages of generators are adjusted to satisfy the small-signal stability constraint, while the outputs of generators are almost unchanged. SCI mainly depends on active power adjustment to improve the small-signal stability, which is also a commonly used method in practical applications. However, this method may cause a large economic loss.

The eigenvalues of the BM and DSCE are shown in Fig. 6. With the proposed method, the objective eigenvalue (whose damping ratio is the smallest) moves from the right side to the left side, which enhances the small-signal stability of the system. To further validate the solutions of DSCE, we set a three-phase short circuit on bus 21 at 1s and clear it at 1.1s. The time-domain simulations for DSCE and BM are shown in Fig. 7 and Fig. 8.

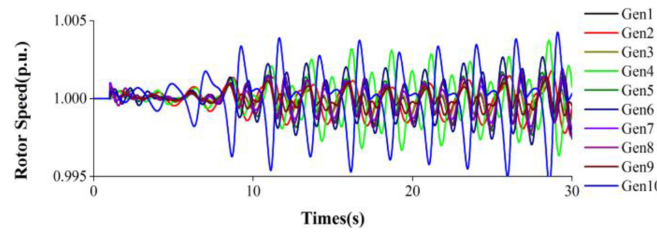


Fig. 7. Time-domain simulation of IEEE 39-bus system for BM.

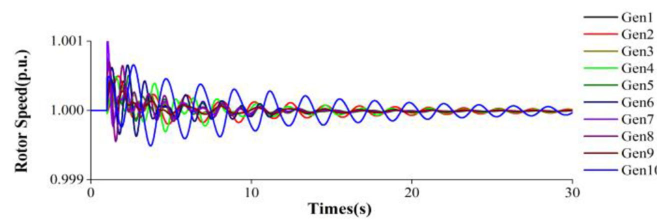


Fig. 8. Time-domain simulation of IEEE 39-bus system for DSCE.

Based on the performance of rotor speeds of all generators, we can conclude that under BM (Fig. 7) the system is unstable while under DSCE (Fig. 8), the oscillation can be quickly damped out.

B. Effectiveness of Error Compensation Strategy

Without the proposed compensation strategy, the recall of the insecure samples R is demonstrated in Table VII. The performance of SVM is shown in Fig. 9. The black line is the visualization of the linear kernel function of SVM, which is used to establish the small-signal stability constraint of the proposed method. We can see that in the IEEE 118-bus system, some insecure samples (i.e., red + in Fig. 9) are misclassified as secure samples (i.e., green + in Fig. 9). With the proposed compensation strategy, the misclassification of insecure samples is avoided, which demonstrates its usefulness to ensure that the re-dispatch result is stable.

C. Comparisons of Computational Efficiency

Matpower is used to solve the proposed OPF model with an interior point solver. The adopted computing platform is a personal workstation with a 3.2GHz Intel Core i5-4460 CPU and 16GB RAM. The runtime environments are MATLAB 6.0 and DIgSILENT/PowerFactory 15.2.

The computing times for eigenvalue and sensitivity calculations of SCI and DSCE are compared in Table V. The sensitivity of SCI is obtained by a numerical method, which is also used by DSCE. SCI requires the calculation of the eigenvalue at each iteration when solving OPF. In DSCE, finding v_{ge} requires a few iterative calculations of sensitivity. In particular, only 1 and

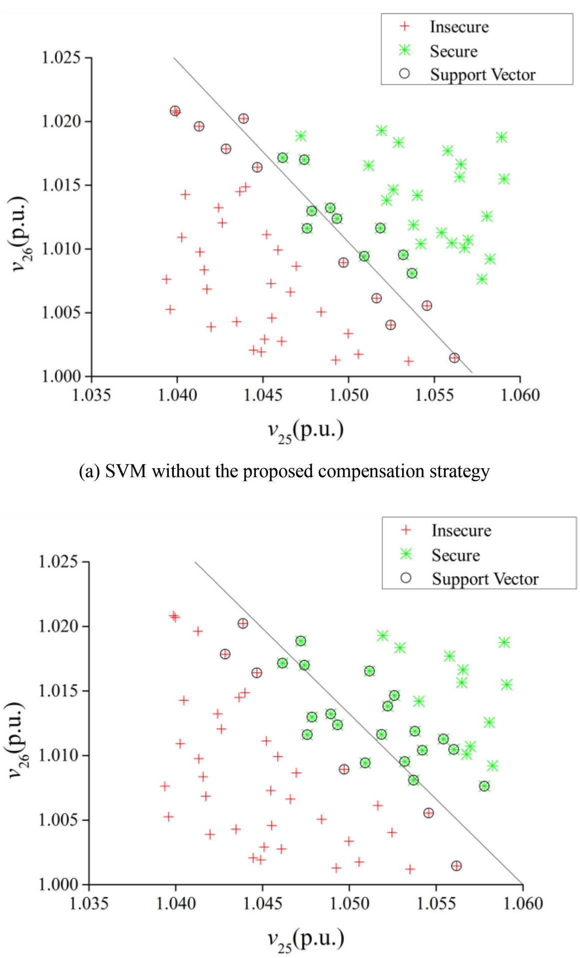
TABLE VIII
COMPUTING TIMES OF DIFFERENT METHODS FOR EIGENVALUE AND SENSITIVITY CALCULATIONS

Calculation step	IEEE 39-bus system					IEEE 118-bus system				
	CPU(s) /call	calls SCI	calls DSCE	Total CPU(s) SCI	Total CPU(s) DSCE	CPU(s) /call	calls SCI	calls DSCE	Total CPU(s) SCI	Total CPU(s) DSCE
Eigenvalue	0.158	54	3	8.53	0.47	0.935	33	5	30.86	4.675
Sensitivity	0.156	486	9	75.82	1.40	0.945	594	54	561.33	51.03

*The sample generation can be achieved by parallel computing. This process counts one call with the longest computational time.

TABLE IX
SIMULATION RESULTS OF DIFFERENT SCALES OF SAMPLING INTERVAL AND NUMBER OF VARIABLES

Number of variables	Complete space				ε_1=0.02p.u.				ε_1=0.01 p.u.				ε_1=0.005 p.u.			
	Cost ($/h)	ζ_m(%)	l	m	Cost ($/h)	ζ_m(%)	l	m	Cost ($/h)	ζ_m(%)	l	m	Cost ($/h)	ζ_m(%)	l	m
2	-	-	-	-	171022	5.81	5	1	170974	5.27	5	1	171050	5.16	3	1
5	-	-	-	-	171016	5.83	1	1	170958	5.03	5	1	171044	5.08	10	1
19	-	-	-	-	-	-	-	-	170994	5.50	-	2	171044	5.08	3	1



(a) SVM without the proposed compensation strategy

(b) SVM with the proposed compensation strategy

Fig. 9. The diagram of SVM performance.

TABLE X
SIMULATION RESULTS OF DIFFERENT SAMPLE SIZE

Sample size	Generation Cost($/h)	ζ_m(%)	l	m
30	170970	5.27	2	1
50	170964	5.13	1	2
100	170974	5.27	5	1
150	170956	5.03	7	1
200	170973	5.27	6	1

4 iterations are needed for the IEEE 39-bus and 118-bus systems, respectively. Note that, as mentioned earlier, only the selected variables ($v_{g,j}$, $j\in\boldsymbol{A}_v$) require sensitivity calculation. In each iteration to find v_{ge}, the eigenvalue will be calculated once. Then, after OPF, another eigenvalue calculation will be conducted to check the system's small-signal stability. In these cases, the small-signal stability requirement can be satisfied within one optimization process. We generate 100 samples to establish the data-driven surrogate constraint. With parallel computing, this process takes the same time needed for calculating the eigenvalue for a single time. DSCE can significantly reduce the number of iterations to speed up the process.

In addition to the calculation of the eigenvalue and sensitivity, the computing times also include SVM and the optimization process. SVM, including penalty factor optimization, takes 0.0057s and 0.0082s for IEEE 39-bus and 118-bus systems, respectively. The optimization process is conducted by Matpower with a user-defined linear inequality constraint and spends 0.1304s and 0.2291s for 39-bus system and 118-bus systems, respectively. In Table VIII, it can be seen that DSCE has approximately 40 and 10 times higher computational efficiency for 39-bus and 118-bus systems compared to SCI, respectively. This demonstrates the high computational efficiency for online applications thanks to the data-driven surrogate constraint modeling.

TABLE XI
SIMULATION RESULTS FOR CONSTRAINT GENERATED BY ELM

Number of variables	ε_1=0.02p.u.			ε_1=0.01 p.u.			ε_1=0.005 p.u.		
	Cost ($/h)	ζ_m(%)	m	Cost ($/h)	ζ_m(%)	m	Cost ($/h)	ζ_m(%)	m
2	170980	5.44	2	170957	5.06	2	171050	5.16	1
5	170970	5.35	3	170973	5.24	2	171114	5.02	3
19	171179	5.25	12	172098	5.56	15	171072	5.14	5

D. Impacts of the Number of Variables, Sampling Space, and Sample Size

Taking the IEEE 118-bus system as an example, we discuss the relationship among the scale of the sampling space, the number of variables, and the sample size. First, we set the sample size to 100 and discuss the impact of the scale of the sampling space and the number of variables. The simulation results are shown in Table IX. If the samples are generated in the whole feasible region, the samples are too unbalanced to create the boundary. With a large sampling space (such as $\varepsilon_1 = 0.02$), the problem of the unbalanced sample may still exist, and the boundary is not approximately linear, which makes the linear kernel difficult to accurately describe it. On the other hand, if the sampling space is too small (such as $\varepsilon_1 = 0.005$), it may miss the optimal set point and lead to some economic loss.

In this paper, according to the discussions in Section III-B, 2 variables are enough for voltage regulation. Other variables can be considered to be redundant. The redundant variables lead to an increased dimension of the sampling space, which may cause the samples to be unbalanced and lead to increased learning difficulty. Without the proposed reduction sampling dimension, the recall of insecure samples cannot be ensured and the proposed method may even be infeasible. Therefore, the number of variables should be as small as possible.

Next, the impact of sample size is discussed. For the case in which $\varepsilon_1 = 0.01$ and 2 variables are chosen, the simulation results are shown in Table X. The samples generated by the Latin hypercube approach are uniformly distributed in the sampling space. For a reasonable scale of sampling space and number of variables, the influence of sample size on the re-dispatch results is limited.

E. Comparison With Other Machine Learning Methods

Taking the extreme learning machine (ELM) [37] as an example, the constraint generated by ELM can be solved by the ipopt solver. The number of neurons is set as 100. The simulation result is shown in Table XI. The convergence of OPF with the constraint generated by ELM is fine. In particular, the performance of the ELM is secure when the number of variables and the scale of the sampling space are small. For large numbers of variables with a large sampling space, ELM cannot fit the boundary with just a small number of samples. The examination strategy needs to be conducted over 10 times, which is not applicable in a practical system. Besides, as ELM cannot accurately fit the boundary, the final costs also increase in the presence of large numbers of variables.

VI. CONCLUSION

This paper proposes a data-driven SSSC-OPF method. A sample generation strategy is proposed to avoid the huge burden of data generation and storage. This is achieved by sampling space compression by reducing the number of variables and compressing the sampling interval via sensitivity analysis. The data-driven surrogate constraint for small signal stability is formulated by SVM leveraging the compressed sampling space. The learned data-driven surrogate constraint, which is explicit and convex, is embedded in the OPF model. An error compensation strategy with penalty factor optimization is proposed for insecure samples. An examination strategy is also proposed to compensate for the error caused by SVM approximation and guarantee the small-signal stability of re-dispatch results. Simulation results on the IEEE 39-bus system and 118-bus systems show that the proposed method is effective in ensuring small-signal stability with a slight compromise on economy. Additionally, the proposed method has a very high computational efficiency for online implementation.

It is worth noting that this paper focuses on single-time scale economic dispatch. It can be extended to multiple-time scale optimization by following the idea in [38]. Besides, the voltage controller is usually implemented in the discrete form [39]. This characteristic will be considered in our future work.

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